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NMAM INSTITUTE OF TECHNOLOGY, NITTE

(An Autonomous Institution affiliated to VTU, Belagavi)

Third Semester B.E. (E&C) (Credit System) Degree Examinations

Supplementary Examinations - December 2025

21EC305 / 20EC305 / 17EC305 - SIGNALS AND SYSTEMS

Duration: 3 Hours

Max. Marks: 100

Note: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II** each and **One full** question from **Unit – III**.

Unit – I

Marks BT* CO* PO*

1. a) Define even and odd signals. Find odd and even components of the following signals.

i) $x(t) = 1 + t \cos t + t^2 \sin t + t^3 \cos^2 t \sin t$ ii) $x(n) = 3n(n-2)$

8 L*3 1 1

- b) For the signal shown in Figure Q1(b) plot

i) $x(-3t+2)$ ii) $x(2(t-2))$ iii) $x(-2t-1)$

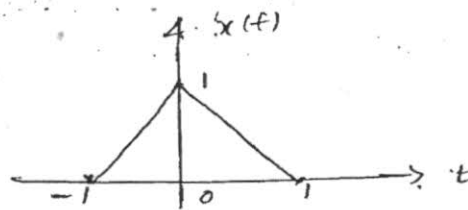


Figure Q1(b)

- c) Find energy or power of the signals

i) $x(n) = \{3, 4, 1, 1, 5\}$

6 L3 1 1

ii) $x(t) = A \cos(W_0 t + \phi)$

6 L3 1 1

2. a) Consider an LTI system with unit impulse response $h(t) = e^{-t} u(t)$. If the input applied to the system is $x(t) = e^{-3t} [u(t) - u(t-2)]$. Find the output of the system and also sketch the response.

8 L3 2 1

- b) Evaluate the discrete time convolution sum given below $xy(n) = \alpha^n u(n) * u(n)$; $0 < \alpha < 1$. Sketch the $y(n)$.

4 L3 2 1

- c) Construct the following signals.

(i) $x(t) = r(t+1) - r(t) + r(t-1)$

(ii) $x(n) = [8 - n] [u(n) - u(n-5)]$

8 L3 2 1

3. a) Prove that for a stable discrete system, impulse response should be absolutely summable.

6 L2 2 1

- b) Find whether following LTI systems are stable, causals, memory less (i) $h(t) = e^{2t} u(t-1)$ (ii) $h(n) = n \left(\frac{1}{2}\right)^n u(n)$

8 L3 2 1

- c) Find the natural response for the system described by the following difference equation.

$y(n) - \frac{9}{16}y(n-2) = x(n-1)$ with $y(-1) = 1$ and $y(-2) = -1$

6 L3 2 1

Unit – II

4. a) Evaluate the Fourier series representation for the signal $x(t) = \cos 4t + \sin 8t$.

6 L3 3 1

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- b) Find the Fourier series co-efficients for the signal $x(t)$ shown in Figure Q4(b)

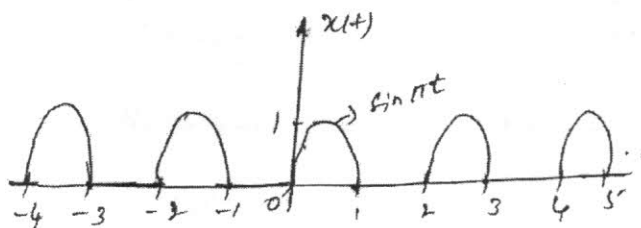


Figure Q4(b)

- | | | | | | |
|----|---|---|----|---|---|
| c) | Specify the Nyquist rate for each of the following signals. | 8 | L3 | 3 | 1 |
| | (i) $x(t) = \sin^2(200t)$ (ii) $x_3(t) = \cos(150\pi t) \sin(100\pi t)$ | 6 | L3 | 3 | 1 |
5. a) Consider the signal
- $$x(n) = 2 + 2 \cos \frac{\pi}{4} n + \cos \frac{\pi}{2} n + \frac{1}{2} \cos \frac{3\pi}{4} n.$$
- (i) Determine and sketch its power density spectrum
(ii) Evaluate the power of the signal.
- | | | | | | |
|--|--|---|----|---|---|
| | | 8 | L3 | 4 | 1 |
|--|--|---|----|---|---|
- b) Find the DTFT of the signal
 $x(n) = \alpha^n u(n)$; $|\alpha| < 1$. Draw the magnitude spectrum.
- | | | | | | |
|--|--|---|----|---|---|
| | | 6 | L3 | 4 | 1 |
|--|--|---|----|---|---|
- c) Find the frequency and impulse response of the system described by the differential equation.
- $$\frac{d^2 y(t)}{dt^2} + \frac{5 dy(t)}{dt} + 6y(t) = -\frac{dx(t)}{dt}$$
- | | | | | | |
|--|--|---|----|---|---|
| | | 6 | L3 | 4 | 1 |
|--|--|---|----|---|---|
6. a) Evaluate the Fourier transform for the signal $x(t) = e^{-3t} u(t-1)$. Also find the expression for magnitude and phase spectra.
- | | | | | | |
|--|--|---|----|---|---|
| | | 8 | L3 | 4 | 1 |
|--|--|---|----|---|---|
- b) Find the inverse Fourier transform of
- $$X(j\omega) = \frac{5j\omega + 12}{(j\omega)^2 + 5j\omega + 6}$$
- | | | | | | |
|--|--|---|----|---|---|
| | | 6 | L3 | 4 | 1 |
|--|--|---|----|---|---|
- c) Determine the DTFS of the signal
 $x(n) = \cos(\frac{\pi}{3})n$ and draw its spectrum
- | | | | | | |
|--|--|---|----|---|---|
| | | 6 | L3 | 4 | 1 |
|--|--|---|----|---|---|

Unit – III

7. a) For the signal
 $x(n) = 7(\frac{1}{3})^n u(n) - 6(\frac{1}{2})^n u(n)$. Find the z-transform and ROC.
- | | | | | | |
|--|--|---|----|---|---|
| | | 8 | L3 | 5 | 1 |
|--|--|---|----|---|---|
- b) List any six properties of ROC.
- | | | | | | |
|--|--|---|----|---|---|
| | | 6 | L2 | 5 | 1 |
|--|--|---|----|---|---|
- c) Using partial fraction expansion method obtain the time domain signal corresponding to the z-transform given below
- $$X(z) = \frac{1+2z^{-1}+z^{-2}}{1-\frac{3}{2}z^{-1}+\frac{1}{2}z^{-2}} \text{ with ROC } |z| > 1.$$
- | | | | | | |
|--|--|---|----|---|---|
| | | 6 | L3 | 5 | 1 |
|--|--|---|----|---|---|
8. a) Using the appropriate properties find the Z-transform of
 $x(n) = (\frac{1}{2})^n u(n) * (\frac{1}{3})^n u(n)$. Also sketch ROC of the same.
- | | | | | | |
|--|--|---|----|---|---|
| | | 8 | L3 | 5 | 1 |
|--|--|---|----|---|---|
- b) Find the input $x(n]$ of the system if $y(n) = (\frac{1}{3})^n u(n) + \frac{2}{3}(-\frac{1}{2})^n u(n)$
and $h(n) = (\frac{1}{2})^n u(n)$.
- | | | | | | |
|--|--|---|----|---|---|
| | | 8 | L3 | 5 | 1 |
|--|--|---|----|---|---|
- c) Determine the inverse Z-transform of $x(z) = \frac{1}{1-z-3}$
- | | | | | | |
|--|--|---|----|---|---|
| | | 4 | L3 | 5 | 1 |
|--|--|---|----|---|---|